

10 (a) (i) change in flux linkage $= 40 \times (5.0 - 3.0) \times 10^{-6}$
 $= 8(0) \times 10^{-5} \text{ Wb}$ A1 [1]

(ii) time taken $= 8.0 \times 10^{-5} / 5.0 \times 10^{-4}$
 $= 0.16 \text{ (s)}$ C1

speed $= 3.0 \times 10^{-2} / 0.16$
 $= 0.19 \text{ (0.188) m s}^{-1}$ A1

or

$E = (\Delta\Phi / \Delta x) \times \text{speed}$
 speed $= 5.0 \times 10^{-4} / (8.0 \times 10^{-5} / 3.0 \times 10^{-2})$ (C1)
 $= 0.19 \text{ (0.188) m s}^{-1}$ (A1) [2]

- (b) a constant non-zero value of E from 0 to 3 cm and
 a different constant non-zero value of E from 3 to 6 cm M1
 E from 3–6 cm has the opposite sign to and larger value than E from 0–3 cm A1 [2]